

PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Computer Engineering Exam

Multiple Choice (1 point each)

Instructions: Choose the best answer for each question.

Question 1: According to the Introduction, what concept was this lab primarily meant to reintroduce?

- A) Kirchhoff's Laws
- B) Ohm's Law
- C) Thevenin's Theorem
- D) Superposition Principle

Question 2: Which of the following PyQt6 modules contains the QMainWindow class?

- A) QtCore
- B) QtGui
- C) QtWidgets
- D) GStreamer

Question 3: According to the cover letter, which technologies is Syed proficient in?

- A) Java, Assembly, and MATLAB
- B) C, C++, Python, and Embedded Linux
- C) HTML, CSS, and JavaScript
- D) Swift, Objective-C, and Xcode

Question 4: In GStreamer with PyQt, which pipeline option is suggested for more flexible uses like overlays and thumbnails, but uses more CPU?

- A) Turning VideoOverlay into a QWidget
- B) "appsink -> QImage -> QLabel/QOpenGLWidget"
- C) Linux X11 sink variant with "ximagesink"
- D) Letting GStreamer decide decode with "decodebin"

Short Answer (2 points each)

Instructions: Answer each question in 2-3 sentences.

Question 5: Briefly explain the objective of the lab described in the 'Objectives' section.

Question 6: What is the primary difference between using QSplitter and QDockWidget for creating user interfaces in PyQt according to the text?

Question 7: Briefly summarize Syed's experience with Toronto MetRobotics based on the cover letter.

Problem-Solving (3 points each)

Instructions: Provide detailed solutions and explanations for each problem.

Question 8: Based on the provided circuit information and experimental setup (Figure 1 and Section 4), describe the steps you would take to verify the output voltage (vO) using simulation software before physically constructing the circuit.

Question 9: Explain how you would implement a feature in PyQt that allows a user to swap the main video feed with a selected thumbnail feed, drawing from the information provided about recreating the current feed title aspect in Qt. Describe the widgets and layouts you would use.

Question 10: Based on Syed's cover letter, describe a scenario where his experience with data pipelines and SQL could be directly applied to improving Geotab's video products. Explain the steps he would take to implement this.